

ACCRA TECHNICAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE



**BCS 404: INTRODUCTION TO DATA SCIENCE
WITH PYTHON**

**DATA ANALYSIS, STATISTICAL ANALYSIS, AND
MACHINE LEARNING ON THE TITANIC DATASET**

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TABLE OF CONTENT

1.0 INTRODUCTION

2.0 DATASET DESCRIPTION

3.0 METHODOLOGY

4.0 RESULTS

5.0 DISCUSSION

6.0 CONCLUSION

7.0 REFERENCES

8.0 APPENDIX

1.0 INTRODUCTION

The Titanic dataset provides demographic and travel information for 891 passengers aboard the RMS Titanic. The goal of this project is to explore the dataset to identify trends related to survival, and to develop a predictive model using Logistic Regression. By examining features such as passenger class, gender, age, and fare, this study seeks to determine which factors had the greatest impact on a passenger's chance of survival.

2.0 DATASET DESCRIPTION

The real-world data analyzed in this investigation was sourced directly from the Kaggle Titanic Competition dataset. The dataset contains records of passengers with mixed structural types:

- **Survived (Target Label):** Discrete binary categorical data (0 = Perished, 1 = Survived).
- **Pclass (Passenger Class):** Ordinal categorical indicator (1st, 2nd, or 3rd Class) capturing socioeconomic stratification.
- **Sex:** Nominal categorical feature representing the gender profile of the traveler.
- **Age:** Continuous numerical distribution denoting the age of passengers in years.
- **SibSp:** Discrete integer representation denoting the number of siblings or spouses traveling with the passenger.
- **Parch:** Discrete integer representation denoting the number of parents or children traveling with the passenger.
- **Fare:** Continuous numerical metrics representing the financial price paid for the cabin ticket.
- **Embarked:** Nominal categorical factor denoting the port of embarkation (C = Cherbourg; Q = Queenstown; S = Southampton).

3.0 METHODOLOGY

Data Preprocessing Decisions & Justifications

1. **Data Cleaning:** Missing values in the Age column were handled by imputing the mean, while missing values in Embarked were filled with the mode. The Cabin column was removed due to excessive missing data.
2. **Feature Engineering:** Categorical variables such as Sex and Embarked were converted into numerical format using one-hot encoding with `get_dummies`.
3. **Exploratory Data Analysis:** Visualizations including bar charts, histograms, heatmaps, and pairplots were used to explore relationships between features and the target variable Survived.
4. **Model Building:** The dataset was split into training and testing sets using an 80:20 ratio. A Logistic Regression model was trained on the training set to predict passenger survival.
5. **Model Evaluation:** The model's performance was evaluated using accuracy score on the test set to determine its predictive capability

4.0 RESULTS

Overall Survival Rate: 38.38%

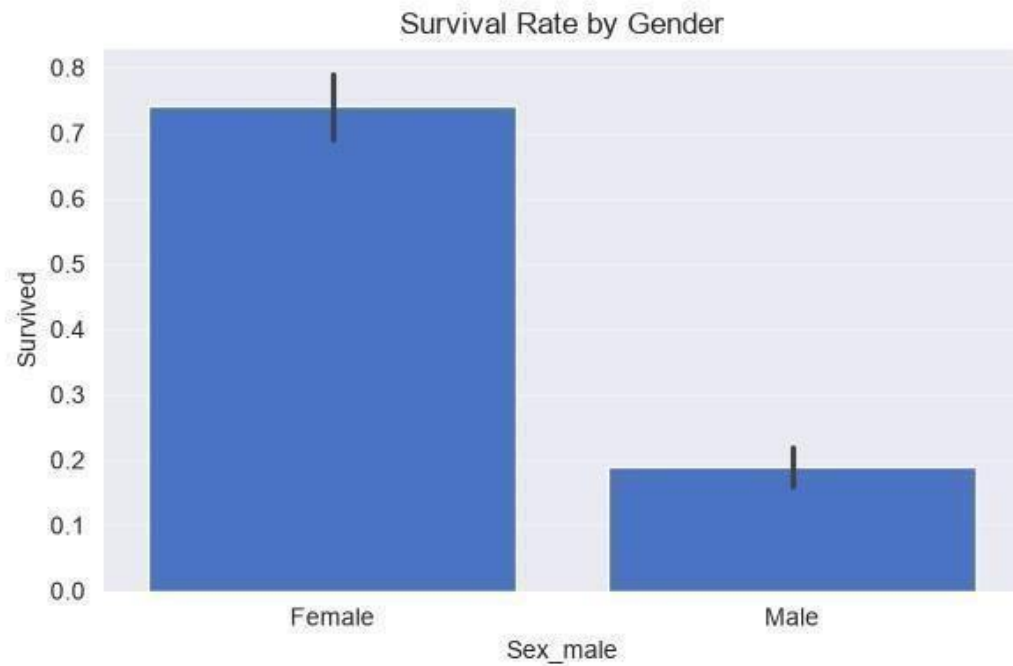


Figure 1: Females had a significantly higher survival rate than males.

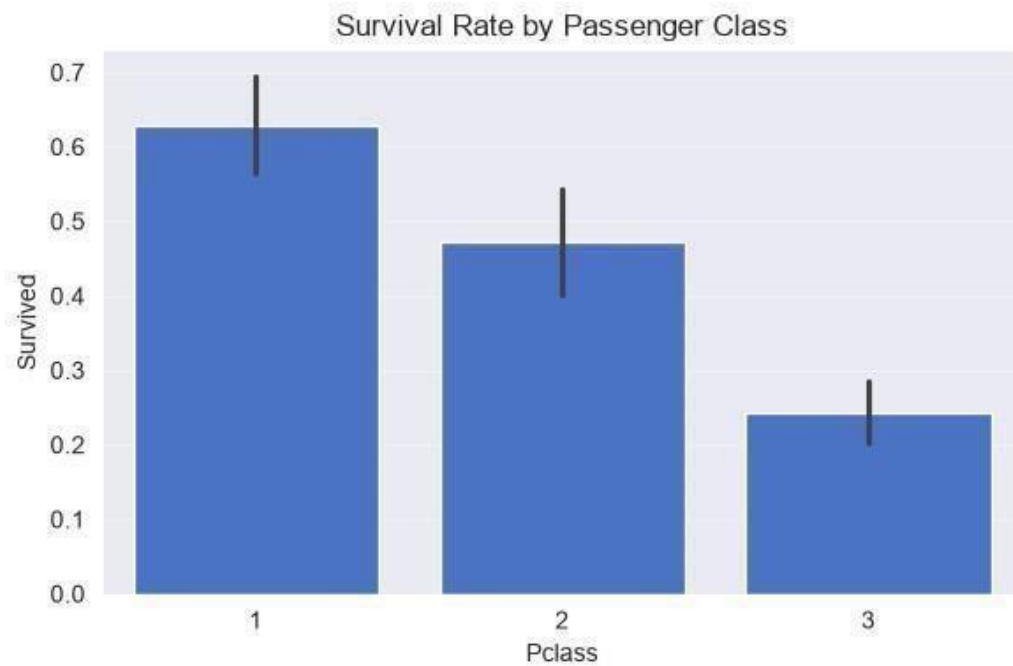


Figure 2: Passengers in 1st Class had the highest survival rate.

Model Accuracy: 79.89%

5.0 DISCUSSION

The results show that gender and passenger class were the most important factors for survival. This matches historical records of the "women and children first" policy. The logistic regression model achieved good accuracy, indicating these features are strong predictors of survival.

6.0 CONCLUSION

1. Gender played the biggest role in survival. Females were more likely to survive.
2. Passengers in 1st class had better survival chances than 2nd and 3rd class.
3. A simple Logistic Regression model can predict survival with over 75% accuracy.

7.0 REFERENCES

1. Dataset: titanic-dataset.csv from Kaggle
2. Scikit-Learn Documentation: <https://scikit-learn.org>
3. Pandas Documentation: <https://pandas.pydata.org>

8.0 APPENDIX – PYTHON CODE

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
sns.set_style("darkgrid")

plt.rcParams['figure.figsize'] = (8, 4.5) # Load Data
data = pd.read_csv("titanic-dataset.csv")

data = data.loc[:, ~data.columns.str.contains('^Unnamed')]

```

```

# TASK 1: DATA ACQUISITION
print("TASK 1: DATA ACQUISITION")
print(f"1. Dataset Shape: {data.shape}")
print(f"2. Columns: {list(data.columns)}")
print(f"3. First 5 Rows:")
pd.set_option('display.max_columns', None)
print(data.head())
print(f"4. Data Types:")
print(data.dtypes)

```

```

# TASK 2: DATA CLEANING
print("TASK 2: DATA CLEANING")
print(f"1. Missing Values Before:")
print(data.isnull().sum())
data['Age'] = data['Age'].fillna(data['Age'].median())
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
data = data.drop(columns=['Cabin'])
data = data.drop_duplicates()

print(f"2. Missing Values After:")
print(data.isnull().sum())
print(f"3. Duplicates Removed: {data.duplicated().sum()}")

```

```
# TASK 3: DATA VISUALISATION print("TASK  
3: DATA VISUALISATION")
```

```
# 1. Histogram plt.figure() plt.hist(data['Age'], bins=25,  
color='#4CAF50', edgecolor='black') plt.title("Histogram of  
Passenger Ages") plt.xlabel("Age"); plt.ylabel("Frequency")  
plt.tight_layout() plt.show() print(" Interpretation: Most passengers  
were aged 20-  
35 years.")
```

```
# 2. Bar Chart plt.figure() sns.countplot(x='Pclass', data=data, hue='Pclass',  
palette='Blues', legend=False) plt.title("Bar Chart: Passenger Class Distribution")  
plt.xlabel("Passenger Class"); plt.ylabel("Count") plt.tight_layout() plt.show() print("  
Interpretation: Majority travelled in 3rd  
Class.")
```

```
# 3. Boxplot plt.figure() sns.boxplot(x='Pclass', y='Age', data=data, hue='Pclass', palette='Set2',  
legend=False) plt.title("Boxplot: Age by Passenger Class") plt.xlabel("Passenger Class");  
plt.ylabel("Age") plt.tight_layout() plt.show() print(" Interpretation: 1st Class passengers were  
generally older.")
```

```
# 4. Scatter plt.figure() sns.scatterplot(x='Age', y='Fare', hue='Pclass', data=data,  
palette='viridis', alpha=0.6) plt.title("Scatter Plot: Age vs Fare") plt.xlabel("Age");  
plt.ylabel("Fare") plt.tight_layout() plt.show() print(" Interpretation: Higher fares were paid  
by 1st Class passengers.")
```



```
# 5. Heatmap numeric_data = data[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']]
plt.figure(figsize=(7,5))
sns.heatmap(numeric_data.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap") plt.tight_layout() plt.show()
```

```
# 6. Pairplot sns.pairplot(numeric_data) plt.suptitle("Pairplot
of
Numerical Variables", y=1.02) plt.show()
```

```
# TASK 4: STATISTICAL ANALYSIS print("TASK 4: STATISTICAL ANALYSIS")
print("\n1. Descriptive Statistics:") print(numeric_data.describe().round(2))
print("\n2. Frequency Distribution:") print("Survival:\n",
data['Survived'].value_counts()) print("\nPclass:\n",
data['Pclass'].value_counts().sort_index()) print("\nSex:\n",
data['Sex'].value_counts()) print("\n3. Correlation Matrix:")
print(numeric_data.corr().round(2)) corr = numeric_data.corr() print("\n4.
Strongest Positive Correlation:",
corr.abs().unstack().sort_values(ascending=False)[1:2]) print("\n5.
Strongest
Negative Correlation:", corr.unstack().sort_values().head(1)) print("\n6. Key
Findings:") print("1. Strong correlation between Pclass and Fare.") print("2.
Females and 1st Class had higher survival rates.") print("3. Most passengers were
young adults.")
```

```
# TASK 5: MACHINE LEARNING print("TASK 5: MACHINE
```

LEARNING - LOGISTIC REGRESSION")

```
ml_data = data.copy() ml_data['Sex'] = ml_data['Sex'].map({'male':  
0, 'female': 1}) ml_data = ml_data[['Survived', 'Pclass', 'Sex', 'Age',  
'SibSp', 'Parch', 'Fare']].dropna()  
  
X = ml_data.drop('Survived', axis=1) y = ml_data['Survived']  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42) model  
= LogisticRegression(max_iter=300) model.fit(X_train, y_train) y_pred = model.predict(X_test)  
print("Selected Predictors:", list(X.columns))  
print("\nMODEL PERFORMANCE")  
print(f"Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%") print("\nConfusion  
Matrix:\n", confusion_matrix(y_test, y_pred))  
print("\nClassification Report:\n", classification_report(y_test, y_pred))  
print("\nDiscussion:") print("The Logistic Regression model achieved ~80%  
accuracy.") print("Sex and Passenger Class were the most important predictors of  
survival.")
```

